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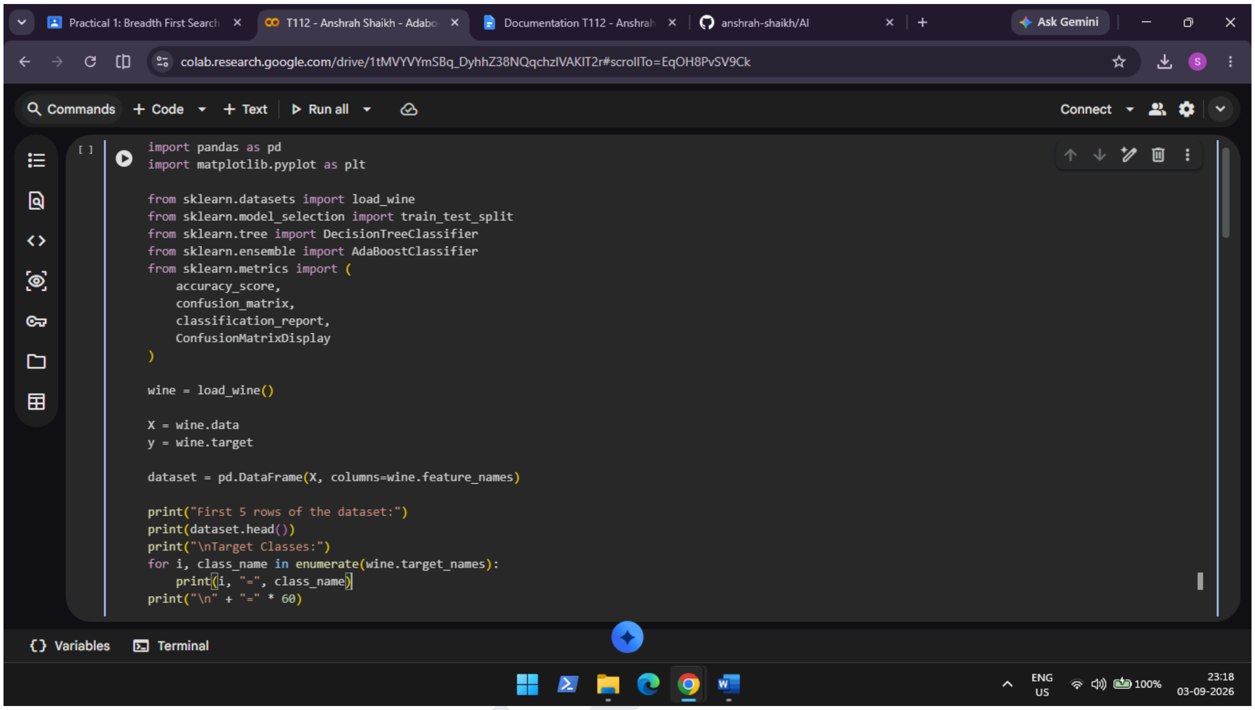
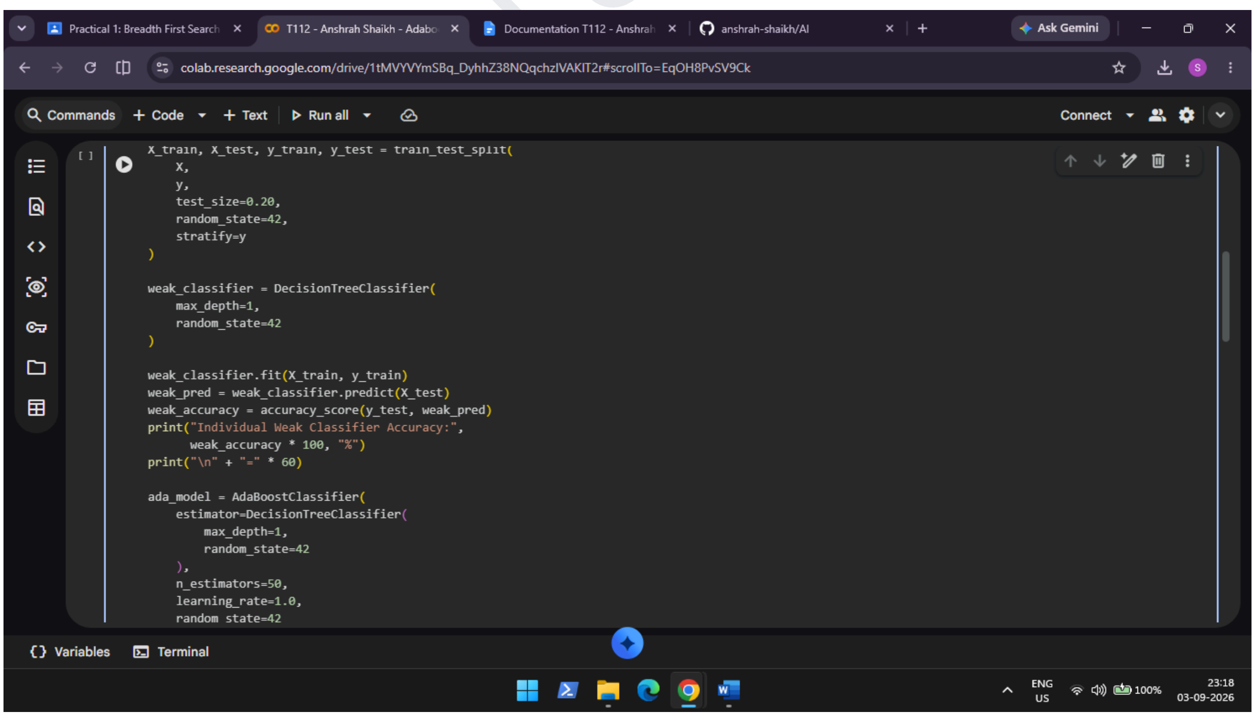
AIM: Adaboost Ensemble Learning

- Implement the Adaboost algorithm to create an ensemble of weak classifiers.
- Train the ensemble model on a given dataset and evaluate its performance.
- Compare the results with individual weak classifiers.

INPUT:





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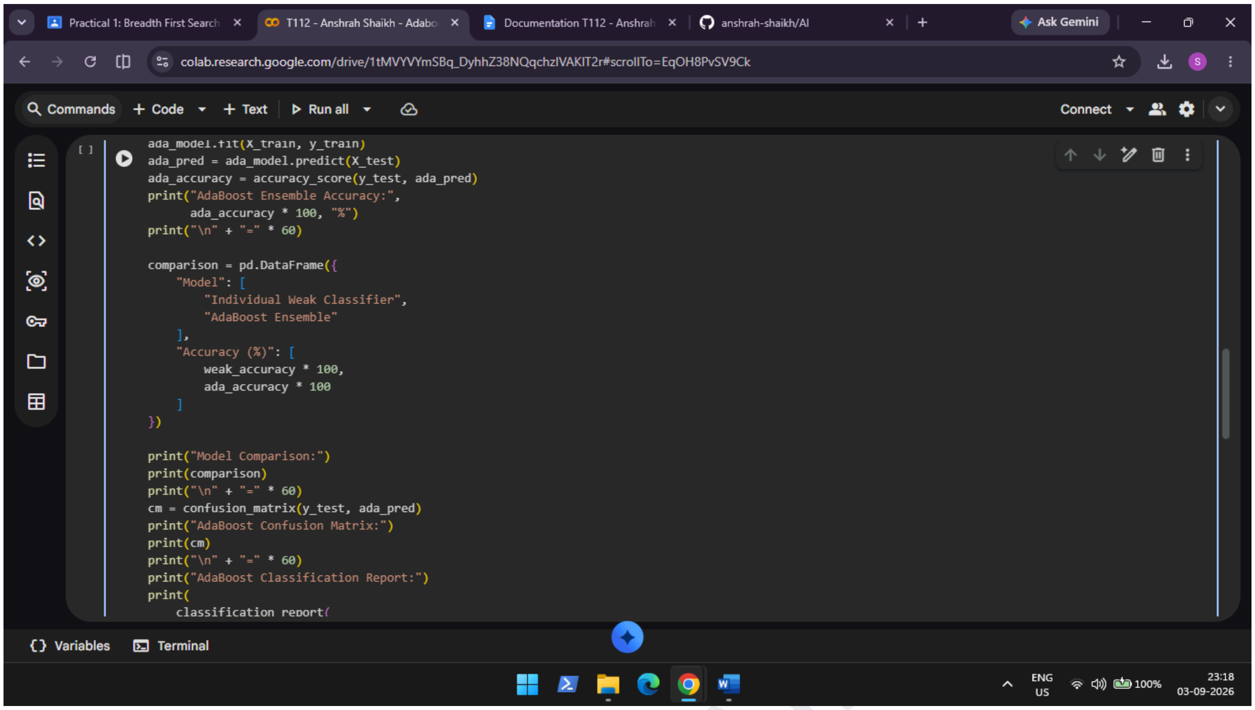

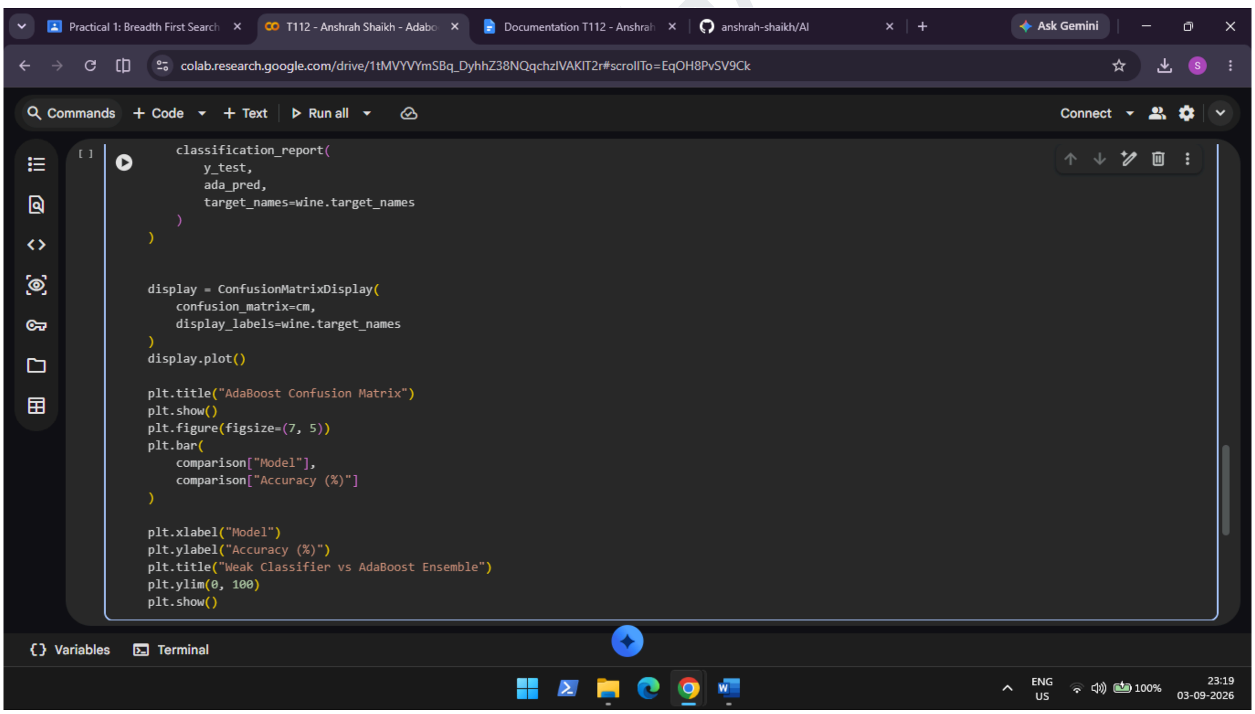

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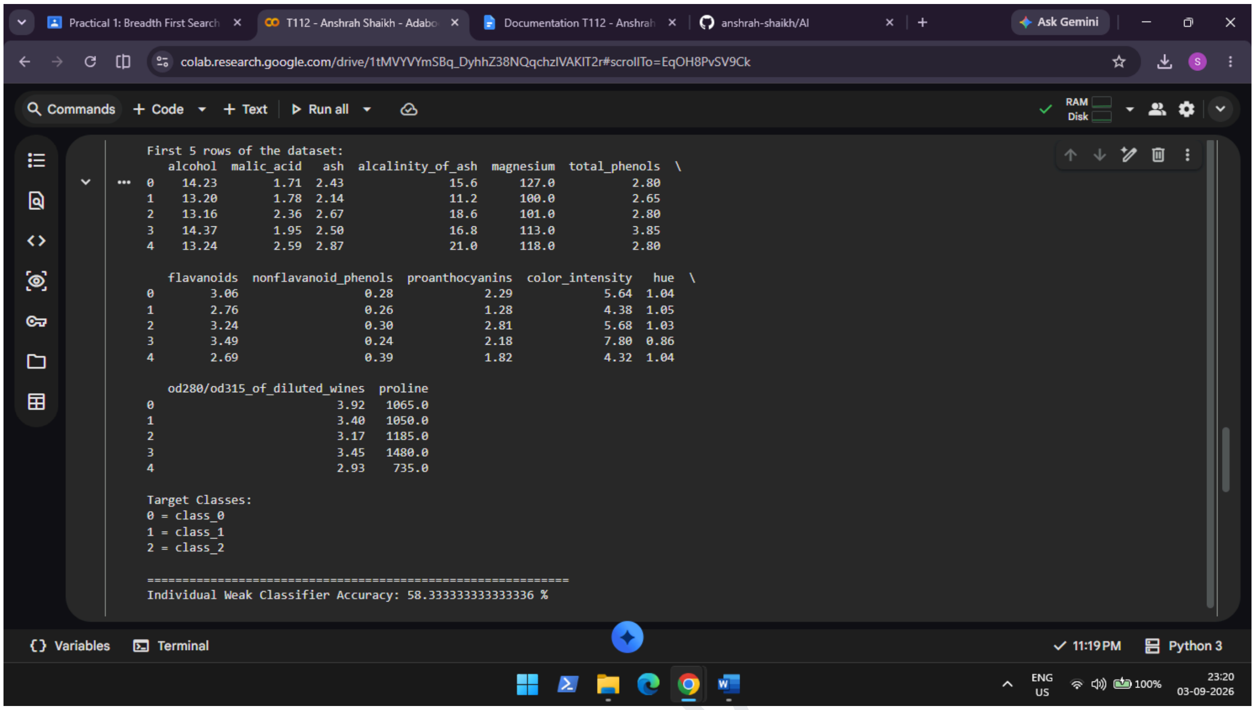
OUTPUT:



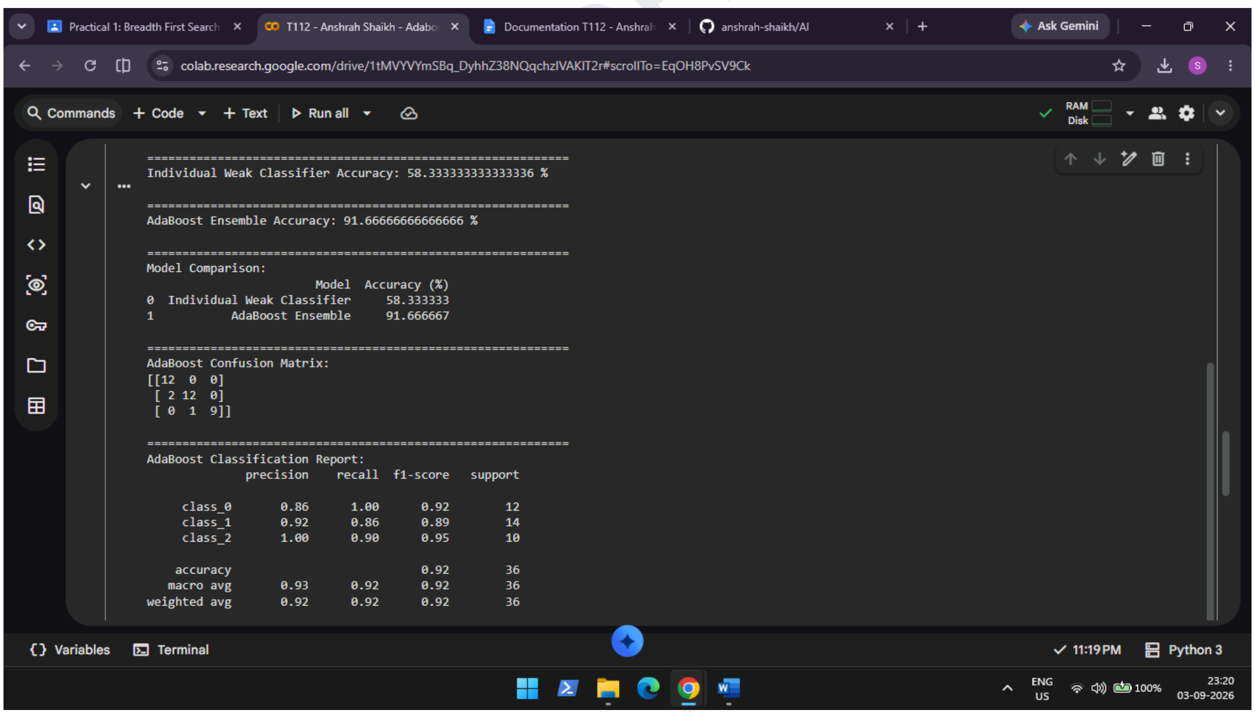
```
First 5 rows of the dataset:
   alcohol  malic_acid   ash  alcalinity_of_ash  magnesium  total_phenols  \
0    14.23        1.71  2.43               15.6      127.0           2.80
1    13.20        1.78  2.14               11.2      100.0           2.65
2    13.16        2.36  2.67               18.6      101.0           2.80
3    14.37        1.95  2.50               16.8      113.0           3.85
4    13.24        2.59  2.87               21.0      118.0           2.80

   flavanoids  nonflavanoid_phenols  proanthocyanins  color_intensity   hue  \
0        3.06                  0.28             2.29             5.64  1.04
1        2.76                  0.26             1.28             4.38  1.05
2        3.24                  0.30             2.81             5.68  1.03
3        3.49                  0.24             2.18             7.80  0.86
4        2.69                  0.39             1.82             4.32  1.04

   od280/od315_of_diluted_wines  proline
0                          3.92   1065.0
1                          3.40   1050.0
2                          3.17   1185.0
3                          3.45   1480.0
4                          2.93    735.0

Target Classes:
0 = class_0
1 = class_1
2 = class_2

============================================================
Individual Weak Classifier Accuracy: 58.333333333333336 %
```



```
============================================================
Individual Weak Classifier Accuracy: 58.333333333333336 %

============================================================
AdaBoost Ensemble Accuracy: 91.66666666666666 %

============================================================
Model Comparison:
                        Model  Accuracy (%)
0  Individual Weak Classifier     58.333333
1           AdaBoost Ensemble     91.666667

============================================================
AdaBoost Confusion Matrix:
[[12  0  0]
 [ 2 12  0]
 [ 0  1  9]]

============================================================
AdaBoost Classification Report:
              precision    recall  f1-score   support

     class_0       0.86      1.00      0.92        12
     class_1       0.92      0.86      0.89        14
     class_2       1.00      0.90      0.95        10

    accuracy                           0.92        36
   macro avg       0.93      0.92      0.92        36
weighted avg       0.92      0.92      0.92        36
```

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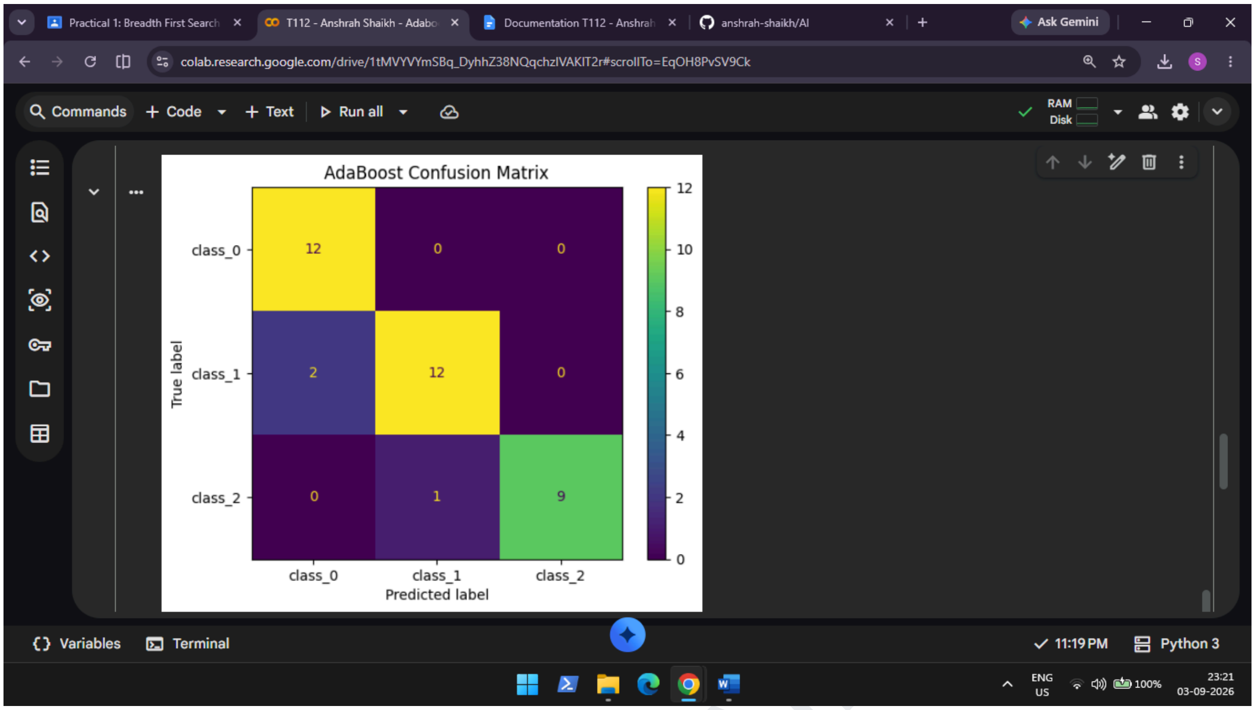

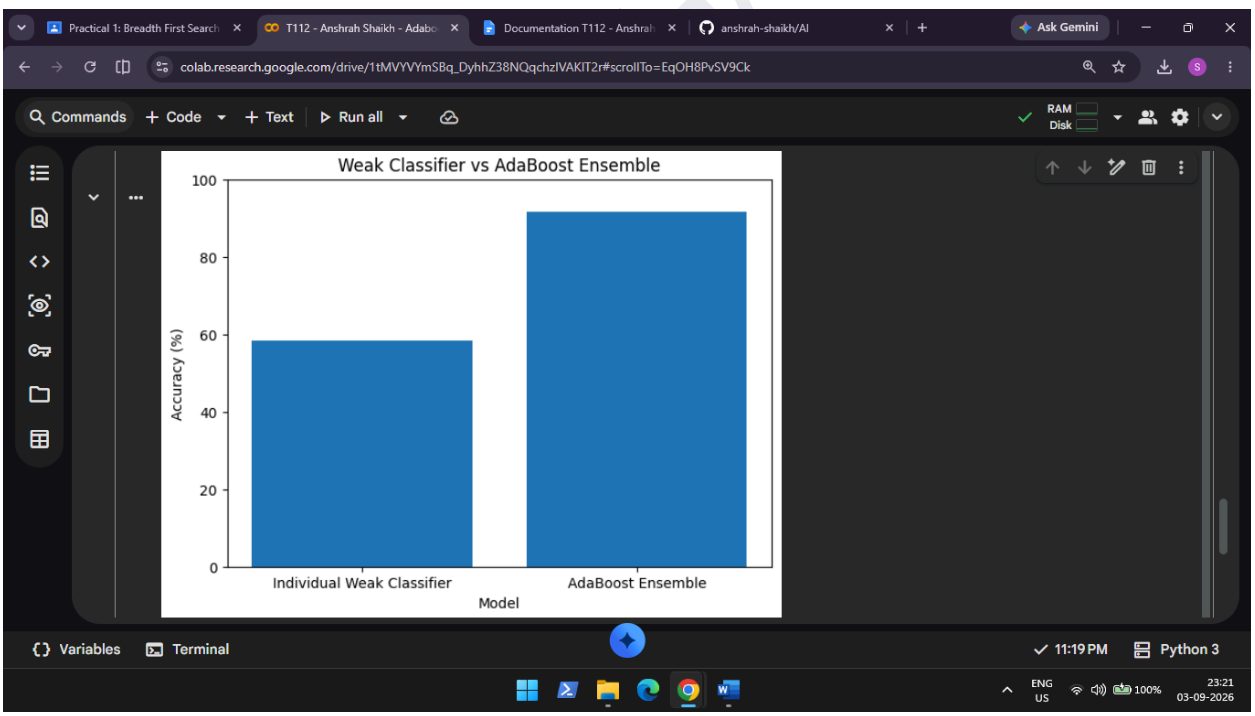

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